ANSWERS

KINEMATICS

1. $\frac{V}{2}$

2. Bead 2

3. (a) $\frac{16}{25}$

   (b) $\sqrt{2}g$

4. (a) $V_{max} = \sqrt{\frac{g^3}{18k}}$

   (b) $X_0 = \frac{3g^2}{16k}$

   (c) $T = \frac{3}{2}\sqrt{\frac{3g}{2k}}$

5. (a) u_0 = 37.5 m/s

   (b) t = 1.5 m/s

6. $\frac{\omega_0}{3}$

NEWTON'S LAWS

1. $\theta = 2\tan^{-1}\left[\left(1+\frac{mg}{F}\right)\cot\alpha\right]$

2. (a) $T = N\cos\theta$

   (b) $\frac{mg}{M\tan\theta + m\cot\theta}$

3. (a) $mg\cos\theta\sqrt{1+\left(\frac{v^2\cos\theta}{Rg}\right)^2}$

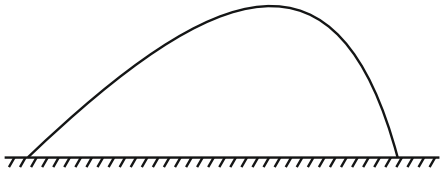
   (b) $\frac{\tan\theta}{\sqrt{1+\left(\frac{v^2\cos\theta}{Rg}\right)^2}}$ where $\tan\theta = \frac{\rho}{2\pi R}$

4. (a) $\vec{a}_{wedge} = \frac{g}{3}\hat{i}$

   $\vec{a}_{x\,block} = \frac{-g}{3}$

   $\vec{a}_{y\,block} = \frac{2g}{3}$;

(b) The block hits the table normally.



(c) $\frac{3u^2}{16g}$

5. (a) $\omega > \sqrt{\frac{g}{l}}$

(b) $\sqrt{\frac{g}{l}} < \omega \leq \sqrt{\frac{2g}{l}}$

(c) $\sqrt{\frac{6g}{l}}$

(d) Parabolic

6. (a) Block with mass M will move up

(b) Yes; (c) $g(1 - \cos\theta)$

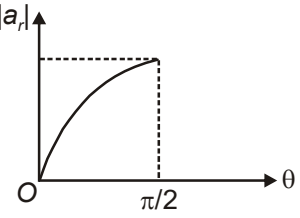
7. (a) $\left(10\sqrt{2} - 8\right)\text{N}$

(b) $\left(5 - 4\sqrt{2}\right)\text{N}$

8. $\frac{3a_0}{2}$

WORK, ENERGY AND POWER

1. (a) $\theta = \cos^{-1}\left(\frac{1}{4}\right)$

(b) 

(c) $t_1 > t_2$

2. (a) $\sqrt{\sqrt{3}gR}$

(b) $mg\left(\sqrt{3} + \frac{1}{2}\right)$

3. (a) $\sqrt{2gR\frac{(M\theta - m\sin\theta)}{M+m}}$

(b) $\frac{mg}{M+m}\left[(M+3m)\sin\theta - 2M\theta\right]$

4. (a) $U = -2mgL + mgd\left(\frac{2-\cos\theta}{\sin\theta}\right)$

   (b) θ = 60°

   (c) Stable

5. The four points are represented by

   $\theta = 0, \pi, \cos^{-1}\left(\frac{u^2-2ag}{3ag}\right)$ and $\left[2\pi - \cos^{-1}\left(\frac{u^2-2ag}{3ag}\right)\right]$

6. (a) 4

   (b) $\frac{1}{2}\sqrt{\frac{g\ell}{2}}$

7. $\theta = \cos^{-1}\left(\frac{2}{3}\right)$

8. $\frac{\left(4+3\sqrt{2}\right)}{2}gl$

9. $\frac{L}{8}\left(2+\sqrt{19}\right)$

MOMENTUM AND CENTER OF MASS

1. (a) 6 cm

   (b) 0.18 m/s

2. (a) $\frac{R}{2\sqrt{2}}$

   (b) $\frac{\left(\sqrt{2}-1\right)R}{2\sqrt{2}}$

3. $v_B = \frac{2\sqrt{31}}{11}u,\ v_C = \frac{4u}{11}$

4. (a) See the solution for graph

   (b) $\sqrt{\frac{n}{n+1}\frac{Fd}{m}}; \sqrt{\frac{Fd}{m}}$

5. $2\sqrt{\frac{Fl}{m}};\ Fl$

6. 45°

7. (b) $\sqrt{u^2+3u_0^2+3uu_0}$

8. (a) $2R + 9h$

ROTATIONAL MOTION

1. 2.26 rad/s

2. $\frac{3\sin\theta}{1+\cos\theta}$

3. 41°

4. (a) $\frac{3F}{4M}$

   (b) $\frac{7F}{8}$

5. A point at a distance $\frac{R}{2}$ from centre

6. (a) 1 m

   (b) 12 m/s

   (c) $3.6\sqrt{3}$ m

7. $v = \sqrt{g\left(a - \frac{b}{2}\right)}$

8. (a) $\frac{v_0^2}{5L}$

   (b) $\frac{3}{2}\frac{v_0}{L}$

9. $\frac{Mg}{2}$

10. (b) $\alpha = \tan^{-1}(2\tan\theta)$

    (c) $\theta_{min} = \tan^{-1}\left(\frac{1}{2\mu}\right)$

    (d) Slippage is more likely when at *A*.

11. $\frac{4g}{11}$; Thread exerts more force on *A*

12. $F = \frac{Idu^2}{2\pi R^4}$

13. $Mg\sqrt{\frac{1}{4} + \frac{1}{\pi^2}}$

GRAVITATION

1. (a) $-\frac{GMm}{243R}$

   (b) $\frac{14045}{14337}\frac{GMm}{R}$

2. $\frac{r_1}{r_2} = \left(\frac{3}{4}\right)^{2/3}$

3. $V_0 = \sqrt{\frac{32}{105}\frac{GM}{R}}$

4. (1)

6. (a) $\left(\frac{\sqrt{7}-2}{6}\right)R$

(b) No

FLUIDS

1. $a = g \tan \theta$ towards right

2. (a) The COM first falls, attains a minimum height and then it rises to original height $\frac{H}{2}$

(b) $x = \left(\sqrt{6}-2\right)H$

3. (a) No

(b) Spring will get compressed more

(c) 100 cm^3

4. (a) ρaL

(b) $h = L\left(1-\frac{a}{g}\right)$

5. (i) $\sqrt{\frac{3}{2}}R$

(ii) (a) $F_v = 1.2 \times 10^7$ N

(b) $F_H = 1.5 \times 10^7$ N

(c) $F = 1.92 \times 10^7$ N

6. (i) $\frac{2A_0 g}{A}$

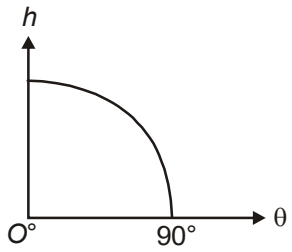
(ii) $v = 2\sqrt{2gH}$

7. $m = \frac{\pi R^3 \rho}{3}$

8. (a) $\rho = \frac{2\rho_0}{3}$

(b) $T = 2\pi\sqrt{\frac{123R}{40g}}$

SURFACE TENSION

1. 

2. (a) $h = \dfrac{2T}{R\rho g}$

   (b) $\dfrac{2T}{\rho g}\left(\dfrac{r+R}{rR}\right)$

3. $\Delta r = \dfrac{4Tr}{3P_0R}$

4. $\dfrac{r_1r_2}{r_1 - r_2}$

5. (a) 120°

   (b) $\sqrt{3}r$

   (c) $\dfrac{3r}{2(2)^{1/3}}$

6. $R = R_0\left[1 - \dfrac{a^4Tt}{2\eta LR_0^4}\right]^{\frac{1}{4}}$

7. (i) $P_o - \rho gh$ (ii) $\sqrt{\dfrac{2T}{\rho g}}$

VISCOSITY

1. $\dfrac{2\eta v_0}{h}$

2. $\dfrac{11}{54}\dfrac{r^2\rho g}{\eta}$ upwards

3. $\dfrac{\eta b\omega R^3}{3t}\left[\left(1+\dfrac{L}{R}\right)^3 - 1\right]$

4. (i) $\pi\rho ga^2\left[\left(\dfrac{b}{a}\right)^2 - 1\right]$ (ii) $Q = \int_a^b v \cdot 2\pi r dr$

5. $\dfrac{2\pi\eta\omega R^3}{3\sin\theta} \simeq \dfrac{2\pi\eta\omega R^3}{3\theta}$

ELASTICITY

1. $x = 1.25\ L$

2. (a) $T = Y\pi r^2\left(\dfrac{d^2}{2l^2}\right)$

   (b) $2T\left(\dfrac{d}{l}\right)$ where T is given in answer (a). No F is much smaller.

4. $h = \dfrac{0.01M\ell}{2m}$

5. $\dfrac{2I\tau_0}{\pi R^4\theta_0}$

SIMPLE HARMONIC MOTION

1. $T = 2\pi\sqrt{\dfrac{LM}{g(M+m)}}$

2. $\omega = \sqrt{\dfrac{8k}{3M}}$

3. (a) $\dfrac{RC}{RD} = \dfrac{2}{1}$

   (b) L.

4. $t = \dfrac{2\pi}{3}\sqrt{\dfrac{L}{3g}}$

5. $f = \dfrac{1}{2\pi}\sqrt{\dfrac{46\,k}{35\,m}};\ A_1 = \dfrac{21}{46}x_0$

6. (a) $2\pi\left(\dfrac{g}{\ell} + \dfrac{k}{m}\right)^{-\frac{1}{2}}$

   (b) $\pi\left[\left(\dfrac{g}{\ell} + \dfrac{k}{m}\right)^{-\frac{1}{2}} + \left(\dfrac{g}{\ell}\right)^{-\frac{1}{2}}\right]$

7. (a) $\dfrac{2\pi}{3}\sqrt{\dfrac{\ell}{g}}$ (b) $2\sqrt{\dfrac{\ell}{g}}\cdot\cos^{-1}\left(\dfrac{\sqrt{19}-1}{6}\right)$

8. (ii) $\omega = \sqrt{\dfrac{k}{m}}$

9. $T = 2\pi\sqrt{\dfrac{L}{2\sqrt{3}g}}$

10. (a) $a\cos\left(\dfrac{5\pi}{18}\right)$

    (b) $\dfrac{5\pi}{9}\sqrt{\dfrac{m}{k}}$

11. (a) 180°, No

    (b) 2 : 1

12. $\dfrac{3\pi}{2}\sqrt{\dfrac{L}{g}}$

13. $\dfrac{20r^2}{L^2}\%;\ 0.05\%$

WAVE MOTION

1. (i) (a) $\sqrt{a_1^2 + a_2^2 + 2a_1a_2\cos(2kx + \delta)}$

   (b) $\frac{A_{max}}{A_{min}} = \frac{a_1 + a_2}{|a_1 - a_2|}$

   (ii) (a) zero (b) zero

2. (a) 98.4 N

   (c) 4.2 Hz decrease.

3. $\frac{\ln 2}{k}$

4. (i) 2

   (ii) 2

5. (a) 380 Hz $\le f \le$ 420 Hz

   (b) $\Delta t_1 > \Delta t_2$

6. (a) $\frac{8}{5}\frac{V}{f_0}$

   (b) $\frac{11}{8}f_0$

8. $v = L\sqrt{\frac{g}{8d}}$

9. (i) (a) $\left|\frac{\partial s}{\partial x}\right|_C = \left|\frac{\partial s}{\partial x}\right|_R$ (b) $\left|\frac{\partial s}{\partial x}\right|_C = \frac{\Delta P}{2B}$ (c) Zero

   (ii) (a) $\frac{0.0004L\,P_0}{3\pi v^2\rho}$ (b) Second

10. (a) x = 4; equal to ρ_0

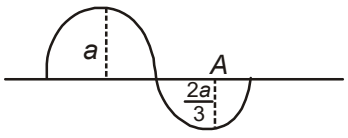
    (b) x = 2, x = 10; $\Delta\rho$ is maximum negative

    (c) x = 4.

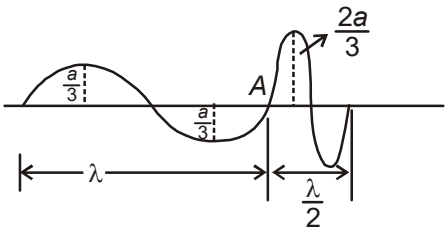
11. (i) 150 m

    (ii) 0.5 s

12. (a)



(b)



TEMPERATURE AND THERMAL EXPANSION

1. $\frac{1}{2}\sqrt{\frac{\alpha_1}{\alpha_2}}$

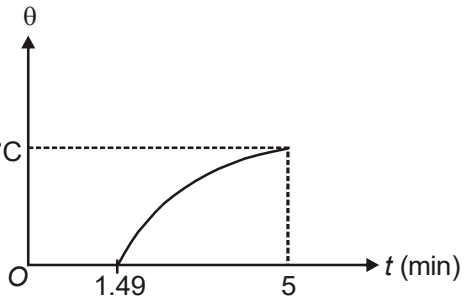
2. $\frac{\alpha_1}{\alpha_2} = \frac{8}{1}$

3. (a) 0.875 *L*
   (b) 0.125 *L*
   (c) Yes

CALORIMETRY

1. 1.71 m

2. (a) Yes
   (b) T_0 = 13.3°C, t_1 = 217 s

3. (a) 22°C
   (b)


KINETIC THEORY OF GASES AND GAS LAWS

1. $u_{rms} = \sqrt{\frac{8}{3}}$

2. (a) 1 (b) $\sqrt{\frac{8kT}{\pi m}}$

3. (a) $v_{rms} = \sqrt{\frac{3}{10}} \times 100$ m/s
   (b) K_T = 15 J

4. $\frac{\sqrt{13}-1}{2}$ m

FIRST LAW OF THERMODYNAMICS

1. (a) $\frac{P_0 M}{\rho_0}\left(\frac{3}{2} + \ln 2\right)$
   (b) $\frac{2}{5}(1 - \ln 2)$

2. (a) $\sqrt{2}$

   (b) 20 second

3. $PV^{1/3}$ = constant; $C = 3R$

4. $V = \dfrac{\sqrt{2}+1}{\sqrt{2}} V_0$

5. (a) $18P_0V_0$

   (b) $\dfrac{19}{2}P_0V_0$

   (c) Answer to (a) does not change. Answer to (b) is $\dfrac{17}{2}P_0V_0$

6. (a) $\gamma = \dfrac{3}{2}$

   (b) $\dfrac{1}{R}\left[\dfrac{15kL^2}{16} + 9P_0AL\right]$

   (c) $\dfrac{27}{16}kL^2 + \dfrac{35}{2}P_0AL$

HEAT TRANSFER

1. $\dfrac{2L^2H}{\left(R^2 - r^2\right)\ell n\left(\dfrac{R}{r}\right)}$

2. $\dfrac{15nRL}{16kA}\ln 2$

3. $\dfrac{10}{11}\lambda_0$

4. (a) 254 K

5. (a) $\dfrac{1}{2}$

   (b) 1

ELECTROSTATICS

1. For both (a) and (b) $W = \dfrac{Qq}{8\pi\varepsilon_0 r}$

2. (a) $\Delta T = \dfrac{KQq}{2\pi R^2}$

   (b) $\dfrac{KQq}{\pi R^2}$

   (c) $\dfrac{KQq}{\pi R^2}$

3. $f = \frac{1}{2\pi}\sqrt{\frac{2g\sin\theta}{x_0}}$

4. (a) $E_0 = \frac{\lambda}{2\pi\varepsilon_0 x}$

   (b) $E_y = \frac{E_0}{2}$

   (c) $E_x = \frac{E_0}{2} = \frac{\lambda}{4\pi\varepsilon_0 x}$

   (d) 45°

5. (a) $\theta = \tan^{-1}\left(\frac{3}{4}\right)$

   (b) Yes

6. (a) $\frac{\sqrt{5}}{6}\frac{\rho R}{\varepsilon_0}$

   (b) $\frac{\sqrt{41}}{24}\frac{\rho R}{\varepsilon_0}$

7. (a) $V = \frac{\sigma}{8\varepsilon_0}\left[\sqrt{R^2 + Z^2} - Z\right]$

   (b) $E_Z = \frac{\sigma}{8\varepsilon_0}\left[1 - \frac{Z}{\sqrt{R^2 + Z^2}}\right]$

8. (a) $r = \frac{16\sqrt{2}\pi\varepsilon_0 R^2 mu^2}{Qq}$

   (b) $V = \sqrt{u^2 + \frac{Qq}{4\pi\varepsilon_0 mR}}$

9. $\frac{1}{4\pi\varepsilon_0}\frac{\lambda}{R}\ln\left(\sqrt{2}+1\right)$

10. $q = 16\pi\varepsilon_0 ar_0$

12. $\frac{1}{4\pi\varepsilon_0}\frac{Q}{R} - \frac{ER}{2}$

13. (a) (i) $E = 0$; (ii) $E = 0$; (iii) $E = \frac{Kq}{x^2}$

    (b) (i) $V = \frac{Kq}{2R}$; (ii) $V = \frac{Kq}{2R}$; (iii) $V = \frac{Kq}{x}$

(c) (i) $V = \frac{Kq}{x^2}$; $V = Kq\left(\frac{1}{R} - \frac{1}{x}\right)$

(ii) $E = 0$; $V = 0$

(iii) $E = 0$; $V = 0$

(d) $-\frac{Kq^2}{2R}$

(e) zero

(f) (i) $E = \frac{Kq}{x^2}$; $V = \frac{Kq}{4R} - \frac{Kq}{x} + \frac{Kq}{R}$ (ii) $E = 0$; $V = \frac{Kq}{4R}$

14. (a) $V = \sqrt{V_0^2 + \frac{\lambda Q}{\pi\varepsilon_0 m}\ln\eta}$

(b) $V_r = \sqrt{V_0^2\left(1 - \frac{1}{\eta^2}\right) + \frac{\lambda Q}{\pi\varepsilon_0 m}\ln\eta}$

15. (a) $\frac{3}{5}\frac{Q^2}{4\pi\varepsilon_0 R}$

(b) $\frac{147}{320}\frac{Q^2}{4\pi\varepsilon_0 R}$

16. (i) at $x = \frac{a}{2}$; $F = \frac{16}{5^{3/2}}\frac{KQP}{a^3}$

at $x = a$; $F = -\frac{KQP}{2^{5/2}a^3}$

(ii) No; Force is zero at $x = \pm\frac{a}{\sqrt{2}}$

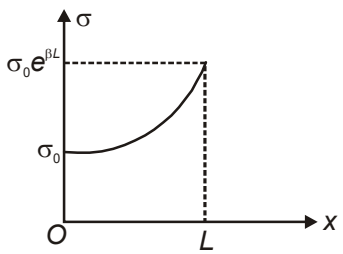
CAPACITOR

1. (a) $\frac{\sigma_{max}}{\sigma_{min}} = \frac{3}{1}$

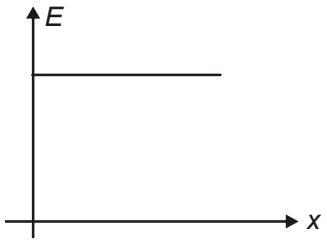
(b) $\sigma_b = \frac{\varepsilon_0 V}{d}(2K - 1)$

(c) $C = \frac{2K\varepsilon_0 A}{d}$

2. (i)



(ii)



(iii) $C = \frac{\varepsilon_0 L}{\beta d}\left[e^{\beta L} - 1\right]$

3. 1.8 mJ

4. (a) $\sigma_1 = \frac{4Q}{L(4L+x)}$; $\sigma_2 = \frac{5Q}{L(4L+x)}$

(b) $\frac{5Qt}{\varepsilon_0 L(4L+x)}$

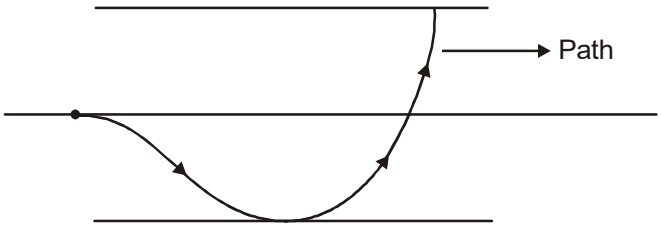
(c) $\frac{4Q}{\varepsilon_0 L(4L+x)}$

(d) $\frac{\varepsilon_0 L(4L+x)}{10t}$

5. (a) $\frac{1}{150}$

(b) 0.67%

6. (a)



(b) $i_0 = \frac{2mgA\varepsilon_0}{q}\sqrt{\frac{g}{3d}}$

7. $\frac{\varepsilon_0 a^2}{d}\left[1 + \frac{a^2\theta^2}{12d^2}\right]$

CURRENT ELECTRICITY

1. (a) $I = \frac{VA}{\rho d}$; $Q = \frac{K\varepsilon_0 AV}{d}$

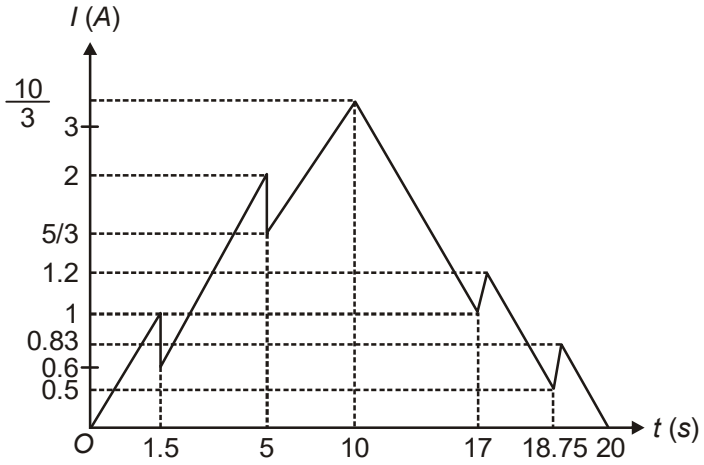
(b) $q = \frac{K\varepsilon_0 AV}{d} e^{-\frac{t}{K\rho\varepsilon_0}}$

2. $Q = (\rho_2 - \rho_1)\,\varepsilon_0 I$

Q is negative if $\rho_1 > \rho_2$

3. (a) $\frac{8R}{15}$

(b) $\frac{3R}{5}$

4. 

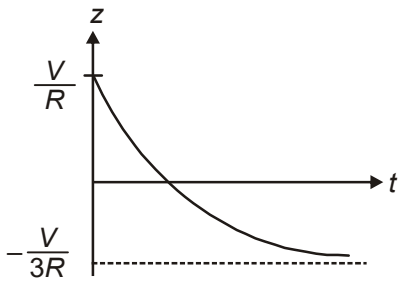
5. (a) $\frac{V}{(8\pi\varepsilon_0 a)R}e^{\frac{-3t}{8\pi\varepsilon_0 aR}}$

(b) $\frac{4\pi\varepsilon_0 aV^2}{3}$

6. (a) $\frac{V}{R}$

(b) $-\frac{V}{3R}$

(c) $Z = \frac{4V}{3R}\left[e^{-\frac{3t}{RC}} - \frac{1}{4}\right]$



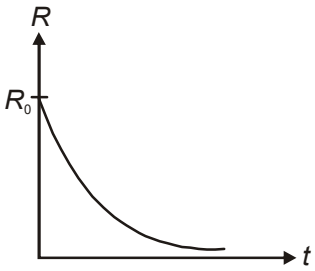
(d) $t_0 = \frac{2RC}{3}\ell n2$

MOTION OF CHARGE IN ELECTROMAGNETIC FIELD

1. 60°

2. (a) $\frac{mv_0}{2qB_0}$

(b)



3. (a) $d < \frac{mv}{qB}(1 - \sin\theta)$

(b) 15°

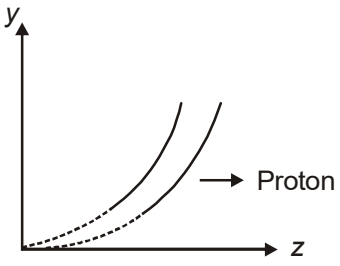
(c) $\pi - 2\theta$

4. $f = \frac{mV}{qB_0\Delta x}$

5. (a) $y = \frac{qE\ell D}{mV_x^2}$

(b) $y = \frac{D\ell qB}{mV_x}$

(c)



6. $R = \frac{mE_0}{qB_0^2}$

MAGNETIC EFFECT OF CURRENT

1. (i) $\frac{\mu_0 NIR^2}{2}\left[\frac{1}{\left(R^2 + x^2\right)^{3/2}} + \frac{1}{\left[R^2 + \left(R - x^2\right)^{3/2}\right]}\right]$

(ii) The magnetic field is constant at points close to mid way between the coils.

2. $\frac{I_2}{I_1} = 1 + \sqrt{2}$

3. Zero

4. (a) $I = \frac{2Mg}{aB}$

(b) $T = \pi\sqrt{\frac{\sqrt{3}a}{g}}$

5. $T_{AO} = Mg + \pi IRB_0$

$T_{BO} = Mg - \pi IRB_0$

ELECTROMAGNETIC INDUCTION

1. (i) $\frac{1}{2} q_0 B_0 \omega_0 L^2$

   (ii) $\frac{1}{2} q_0 \left[B_0 \omega_0 L^2 - \frac{q_0}{C} \right]$

2. (a) $\frac{\mu_0 \pi a^2 \beta}{\ell}$

   (b) $I = I_0 e^{-\frac{2\rho t}{\mu_0 a d}}$

3. $L = \frac{\mu_0 x}{\pi} \ell n \left(\frac{d - a}{a} \right)$

4. No change

5. $\frac{L\alpha}{4} \sqrt{4R^2 - L^2}$

6. (a) $B_z = \frac{\mu_0}{4\pi r^3} [2 \cos^2 \theta - \sin^2 \theta]$

   (b) $\phi = \frac{\mu_0 M R^2}{2\left(R^2 + z^2\right)^{3/2}}$

   (c) $E_{in} = \frac{3\mu_0 M R^2 v z}{2\left(R^2 + z^2\right)^{5/2}}$

7. $\frac{\mu_0 I_0 a}{2\pi R} \ell n 3$

8. (a) Zero

   (b) $\frac{\pi}{4} q a^2 \alpha$

ALTERNATING CURRENT

1. $\frac{\sqrt{2}}{\omega C}$

2. $i = V_0 \left[\frac{1}{R} \sin \omega t + \left(\omega C - \frac{1}{\omega L} \right) \cos \omega t \right]$

3. (a) Source frequency is greater than resonant frequency. Current lags.

   (b) 5 volt

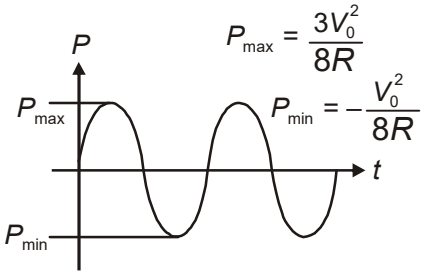
   (c) $V_0 \cos \omega t$

4. (a) $\sqrt{\dfrac{1+\left(\dfrac{\omega L}{R}\right)^2}{1+\left(\dfrac{\omega L}{R}-\dfrac{1}{\omega RC}\right)^2}}$

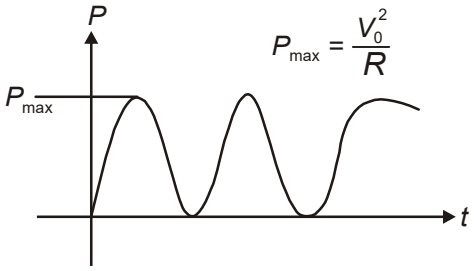
(b) 1

(c) ωRC

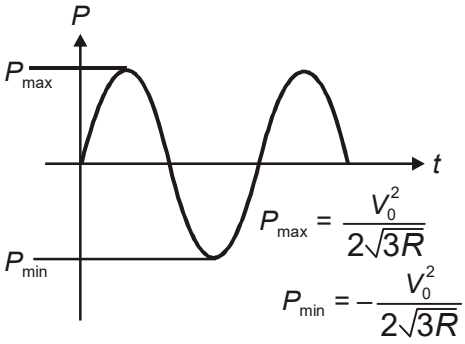
5. (a)



(b)



(c)



GEOMETRICAL OPTICS

1. $f_0\left[\dfrac{4b\Delta\lambda}{\lambda_0\left(a\lambda_0^2+b-\lambda_0^2\right)}\right]$

2. $\dfrac{f}{2}$

3. d = 20 cm

4. $\theta = \dfrac{2r}{5H}$

5. $\mu = \sqrt{\dfrac{R+a}{R-a}}$

6. $\dfrac{f\theta^2}{2}$

WAVE OPTICS

1. $\frac{3}{2}$

2. (a) $\frac{R}{2}$

   (b) $10R$

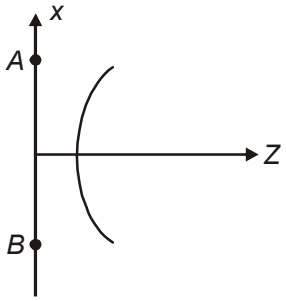
3. $\theta = \sin^{-1}\left[\frac{1}{\mu d}(n\lambda - d\sin\phi)\right]$; $[n = 0, \pm 1, \pm 2 \ldots]$

4. (a) $y' = -\sin \pi t$

   (b) $\frac{1}{6}$ s

5. 9.3 μm

6. Intensity $\propto \cos^2\left(\frac{\pi d_1 d_2}{2\lambda L_1}\right)\cos^2\left(\frac{\pi d_2 y}{\lambda L_2}\right)$

7. 

   Yes the wavefront is getting focussed.

WAVE PARTICLE DUALITY AND ATOMIC PHYSICS

1. 7 W

2. (a) 5536 Å

   (b) 0.089 W

3. 1.33×10^{-7} N

4. $\frac{2P_0}{c}\cos 15°$

5. (a) $\frac{8\pi}{3}\left(\frac{v}{c}\right)^3$

   (b) No, Yes

6. (a) 10^5 volt

   (b) $\lambda_{k\alpha} = 0.23$ Å

   (c) $Z = 74$

7. 20 eV, 13.2 eV

NUCLEAR PHYSICS

1. 0.7820 MeV

2. (a) $\frac{1}{9}$

   (b) 1215 s

3. (a) 8α and 6β

   (b) 4.25×10^9 yr

4. $\left(\frac{M - m_n}{M + m_n}\right)^2$

5. (a) $V_0 = \frac{E_2}{e}$

   (b) Zero

   (c) $q = -\frac{4\pi\varepsilon_0 abE_2}{e(b - a)}$

6. $\frac{2t_0 - \lambda t_0^2}{\lambda^2}$

Answer Key

1. (a) (i) Cl, (ii) Cs, (iii) Al, (iv) F, (v) O, (vi) Xe, (vii) C, (viii) Li, (ix) Br.
   (b) (i) Cl, (ii) Na, (iii) Ar, (iv) Cl, Ar, (v) Al, (vi) Cl, (vii) P, (viii) S.
   (c) (i) Ar, (ii) Cl, (iii) Cl, (iv) Ar, (v) Na.
   (d) (i) A < C < B, (ii) C < B < A, (iii) A < B < C, (iv) C < B < A.
2. (3.726)
3. (4.8)
4. (1.73)
5. (d)
6. (c)
7. (d)
8. (b)
9. (a)
10. (a)
11. (b)
12. (a)
13. (a → p); (b → q); (c → q); (d → r, s)
14. (a → p, q); (b → r); (c → s); (d → s, t)
15. (a) H_2O has a bent structure with HOH bond angle is 104.5°. H_2O_2 has an open book structure with H — O — O angle equal to 94.8° and dihedral angle equal to 111.5° in gaseous phase.
    (b) $AlCl_3$ is a salt of a weak base and a strong acid.
    In normal water, it undergoes hydrolysis.
    In acidic water, H^+ ions react with $Al(OH)_3$ giving $Al^{3+}(aq)$ ions.
    $$AlCl_3(s) \xrightarrow[\text{Water}]{\text{Acidified}} Al^{3+}(aq) + 3Cl^-(aq)$$
    In alkaline water, $Al(OH)_3$ reacts to form soluble tetrahydroxo-aluminate complex or meta-aluminate ions $\left[AlO_2^-(aq)\right]$.
16. (5.6×10^2 g)
17. (300 ppm)
18. (0.2125)
19. (Normality = 0.08, Percentage = 0.136, Volume strength = 0.0448)
20. (c)
21. (a)

22. 'B' is K_2SO_4 and 'C' is $K_2SO_4.\ Al_2(SO_4)_3.24H_2O$

23. (a) Sodium carbonate is first converted into sodium sulphite by passing sulphur dioxide.

$Na_2CO_3 + SO_2 \longrightarrow Na_2SO_3 + CO_2$

The solution of Na_2SO_3 is then boiled with sulphur for two hours.

$Na_2SO_3 + S \longrightarrow Na_2S_2O_3$

(b) It is prepared by dissolving NH_4Cl in disodium hydrogen phosphate in molecular proportion in hot water.

$NH_4Cl + Na_2HPO_4 \longrightarrow NaNH_4HPO_4 + NaCl$

(c) The conversion is made by Solvay process.

$CO_2 + NH_3 + H_2O \longrightarrow NH_4HCO_3$

$NH_4HCO_3 + NaCl \longrightarrow NaHCO_3 + NH_4Cl$

$2NaHCO_3 \longrightarrow Na_2CO_3 + H_2O + CO_2$

(d) The solution of sodium sulphite is boiled with sulphur in an iron retort for two hours.

$Na_2SO_3 + S \longrightarrow Na_2S_2O_3$

(e) Sodium is first converted into sodamide by passing dry ammonia gas over heated sodium.

$2Na + 2NH_3 \longrightarrow 2NaNH_2 + H_2\uparrow$

The liquid sodamide is then dropped over red hot charcoal.

$$2NaNH_2 + C \longrightarrow \underset{\text{Sod. cyanamide}}{Na_2CN_2} + 2H_2$$

$Na_2CN_2 + C \longrightarrow 2NaCN$

24. (i) Black ash is impure sodium carbonate produced in Le-Blanc method when salt cake is reduced by coke.
(ii) Ammonia is liberated
(iii) Fused sodium hydroxide.
(iv) Sodium carbonate decahydrate.
(v) Sodium chloride solution.
(vi) Sodium bicarbonate.
(vii) Sodium.
(viii) Lithium chloride.
(ix) Calcium chloride.
(x) Sodium formate.

25. (i) $2KNO_3 \longrightarrow 2KNO_2 + O_2$

(ii) Water of crystallisation is lost and the substance becomes anhydrous,

$Na_2CO_3.10H_2O \longrightarrow Na_2CO_3 + 10H_2O$

(iii) $$NaNH_4HPO_4 \longrightarrow \underset{\substack{\text{Sodium}\\\text{metaphosphate}}}{NaPO_3} + NH_3 + H_2O$$

(iv) $2NaHCO_3 \longrightarrow Na_2CO_3 + H_2O + CO_2$

(v) $2LiOH \longrightarrow Li_2O + H_2O$

(vi) $2LiNO_3 \longrightarrow Li_2O + 2NO_2 + \frac{1}{2}O_2$

(vii) $Li_2CO_3 \longrightarrow Li_2O + CO_2$

(viii) $4Na_2S_2O_3 \longrightarrow 3Na_2SO_4 + Na_2S_5$

26. (i) When heated with NaOH, NH_4Cl evolves ammonia gas.

$NH_4Cl + NaOH \longrightarrow NH_3 + NaCl + H_2O$

(ii) On treatment with dil. H_2SO_4, a pungent gas, SO_2 is evolved which turns acidified $K_2Cr_2O_7$ green.

$Na_2SO_3 + H_2SO_4 \longrightarrow Na_2SO_4 + H_2O + SO_2$

(iii) On addition of $AgNO_3$ solution, sodium thiosulphate gives a white precipitate which turns black.

$2AgNO_3 + Na_2S_2O_3 \longrightarrow Ag_2S_2O_3 + 2NaNO_3$

$Ag_2S_2O_3 + H_2O \longrightarrow Ag_2S + H_2SO_4$

(iv) On addition of $AgNO_3$ solution, KI gives a yellow precipitate of AgI.

$2AgNO_3 + 2KI \longrightarrow 2AgI + 2\ KNO_3$

(v) On addition of dil. HCl, Na_2SO_3 gives pungent gas which turns acidified $K_2Cr_2O_7$ paper green.

(vi) Flame test.

27. (B) is H_2O and (A) is CO_2. X is $KHCO_3$

$$\underset{(X)}{2KHCO_3} \longrightarrow \underset{(Y)}{K_2CO_3} + \underset{(A)}{CO_2} + \underset{(B)}{H_2O}$$

28. X is $NaHCO_3$.

$$\underset{(X)}{2NaHCO_3} \longrightarrow \underset{(Y)}{Na_2CO_3} + \underset{(A)}{CO_2} + \underset{(B)}{H_2O}$$

29. (d)

30. (d)

31. (a) Plaster of Paris, $2CaSO_4.H_2O$

(b) Magnesium sulphate, $MgSO_4.7H_2O$

(c) Calcium carbide, CaC_2

$CaC_2 + 2H_2O \longrightarrow Ca(OH)_2 + C_2H_2$

(d) Calcium cyanamide (nitrolim), $CaCN_2$

(e) Calcium hydride (hydrolith), CaH_2

32. Perhaps some of the oxide escaped as smoke or the magnesium did not react completely or the magnesium may have combined with nitrogen to make magnesium nitride instead.

33. molecular formula = $BeN_4Cl_2H_{12}$

The reaction with silver ions suggests that 2 chlorine atoms are present as chloride ions. Thus, the structural formula of the compound is $[Be(NH_3)_4]Cl_2$.

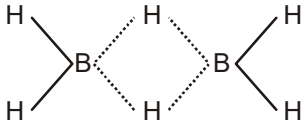
34. (a)

35. (c)

36. XY = $Al_2(SO_4)_3$; [A] = $Al(OH)_3$; [B] = $NaAlO_2$;
[C] = $Al(OH)_3$ [D] = $BaSO_4$

37. $\underset{(X)}{BCl_3}$

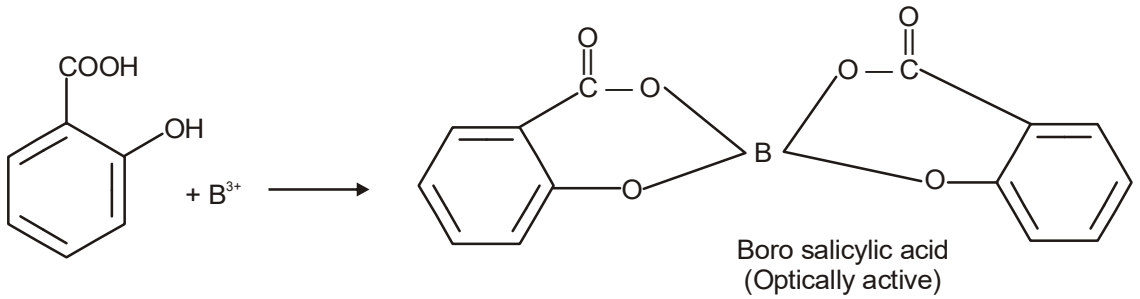
Structure of $\underset{(Y)}{B_2H_6}$



38. (d)

39. (b)

40. Boro salicylic acid is optically active.



Boro salicylic acid
(Optically active)

41. (BA) = $Pb(NO_3)_2$; (X) = NO_2; (Y) = $PbCrO_4$; (Z) = Na_2PbO_2]
42. (A) = $PbCO_3$; (B) = PbO; (C) = Pb_3O_4; (D) = PbO_2
43. (A) is $Pb(CH_3COO)_2$
44. X = Sn; Y = SnS; Z = SnO_2
45. (A) = $SnCl_2$; (B) = Sn(OH) Cl; (C) = $Sn(OH)_2$
46. (b)
47. (a)
48. (c)
49. (i) Quartz glass used in the manufacture of optical instruments is called vitrosil.
    (ii) Carbogen is a mixture of 95% O_2 and 5% CO_2. It is used for artificial respiration for the victims of carbon monoxide poisoning.
    (iii) Kieselguhr — It is a mass of hydrated silica (SiO_2) formed from skeletons of minute plants known as diatoms. It is highly porous and absorbent material used in the manufacture of dynamite.
    (iv) Chrome yellow ($PbCrO_4$) is used as yellow pigment chrome red ($PbCrO_4$.PbO) is used as red pigment.
    (v) $SnCl_4.5H_2O$ is called butter of tin and is used as a mordant in dyeing.
    (vi) Red form of PbO is called litharge and yellow form of PbO is called massicot.
    (vii) Carburated water gas is a mixture of carbon monoxide (23-28%); H_2(34-38%), saturated hydrocarbons (17-21%) unsaturated hydrocarbons (13-16%); CO_2 (upto 2%) and (N_2 upto 5%).
    (viii) Crooke's glass is used in UV filter glasses. It contains CeO_2.
    (ix) Sometimes CaF_2 is added to obtain opal glass.
50. (A) CO; (B) CO_2; (C) H_2O; (D) $COCl_2$; (E) NH_2CONH_2
51. Explanation Based
52. The gas (X) is NH_3.
    It's aqueous solution is NH_4OH which turns red litmus to blue.

$$FeCl_3 + 3NH_4OH \longrightarrow \underset{\text{Brownish ppt.}}{Fe(OH)_3} + 3NH_4Cl$$

$$Fe(OH)_3 + 3HNO_3 \longrightarrow \underset{\text{Soluble}}{Fe(NO_3)_3} + 3H_2O$$

53. (i) NH_4NO_3 (ammonium nitrate)
    (ii) NH_4NO_2 (ammonium nitrite)
    (iii) $Ca_3(PO_4)_2$ (calcium phosphate)
    (iv) NH_3 (ammonia)
    (v) N_3H (hydrazoic acid)
    (vi) N_2H_4 (hydrazine)
    (vii) $(NH_4)_2SO_4$ (ammonium sulphate)
    (viii) NH_2OH (hydroxylamine)
    (ix) N_2O and NO
    (x) H_3PO_4 (orthophosphoric acid)

54. (i) $\underset{(A)}{PH_4I}$ $\underset{(B)}{PH_3}$ $\underset{(C)}{KI}$

(ii) $\underset{\substack{(D)\\ \text{Metaphosphoric Acid}}}{HPO_3}$

(iii) $\underset{(E)}{Cu_2I_2}$

55. (A) = NH_4NO_3, (B) = N_2O, (C) = H_2O, (D) = N_2, (E) = O_2, (F) = NH_3

56. $Bi(NO_3)_3$ undergoes hydrolysis forming nitric acid. Nitric acid liberates iodine from KI which dissolves in excess of KI giving clear yellow solution.

$Bi(NO_3)_3 + H_2O \longrightarrow [Bi(OH)(NO_3)_2] + HNO_3$

$2HNO_3 \longrightarrow H_2O + 2NO_2 + O$

$$2KI + 2HNO_3 + O \longrightarrow 2KNO_3 + \underset{\text{(Dark borwn ppt.)}}{H_2O + I_2}$$

$$I_2 + KI \longrightarrow \underset{\text{Yellow solution}}{KI_3}$$

57. (A) is H_3PO_2, i.e., hypophosphorus acid.

$\underset{\substack{(B)\\ \text{Phosphorus acid}}}{H_3PO_3}$ $\underset{\substack{(C)\\ \text{Orthophosphoric acid}}}{H_3PO_4}$ $\underset{\substack{(D)\\ \text{White ppt.}}}{Hg_2Cl_2\,I}$ $\underset{\substack{(E)\\ \text{Grey ppt.}}}{2Hg\;I}$

58. (a)

59. (a)

60. Explanation Based

61. Substance (A) is sulphur.

$$\underset{(A)}{S} + Na_2SO_3 \longrightarrow \underset{(C)}{Na_2S_2O_3}$$

$$\underset{(C)}{Na_2S_2O_3} + H_2SO_4 \longrightarrow Na_2SO_4 + \underset{(D)\ \text{Turbidity}}{SO_2} + S + H_2O$$

$$2Na_2S_2O_3 + I_2 \longrightarrow \underset{\text{Colourless solution}}{Na_2S_4O_6} + 2NaI$$

62. (a)

63. (a)

64. (b)

65. (d)

66. (b)

67. Compound (X) gives a brick red flame in flame test. Thus, it is a calcium compound. It smells of chlorine on exposure suggests that it is bleaching powder.

68. (i) $HClO < HClO_2 < HClO_3 < HClO_4$

(ii) $HClO < HClO_2 < HClO_3 < HClO_4$

(iii) $HClO_4 < HClO_3 < HClO_2 < HClO$

(iv) $F^- < Cl^- < Br^- < I^-$

(v) $HI < I_2 < ICl < HIO_4$

(vi) $HOI < HOBr < HOCl < HOF$

(vii) $I_2 < Br_2 < Cl_2 < F_2$

(viii) HF < HCl < HBr < HI

(ix) I < Br < Cl < F

(x) I < Br < F < Cl

69. The black powder is MnO_2 and greenish-yellow gas is chlorine.

$MnO_2 + 2NaCl + 3H_2SO_4 \longrightarrow 2NaHSO_4 + MnSO_4 + Cl_2 + 2H_2O$

$3Cl_2 + 8NH_3 \longrightarrow N_2 + 6NH_4Cl$

$3Cl_2 + 6KOH \longrightarrow KClO_3 + 5KCl + 3H_2O$

$2KClO_3 + [MnO_2] \longrightarrow 2KCl + 3O_2 + [MnO_2]$

70. The compound (X) is bleaching powder, $CaOCl_2$.

(i) $CaOCl_2 + 2CH_3COOH + 2KI \longrightarrow Ca(CH_3COO)_2 + 2KCl + H_2O + I_2$

(ii) $CaOCl_2 + CO_2 \longrightarrow CaCO_3 + Cl_2$

(iii) $Ca(OCl)Cl + H_2O \longrightarrow Ca(OH)_2 + Cl_2$

$C_2H_5OH + Cl_2 \longrightarrow CH_3CHO + HCl$

$CH_3CHO + 3Cl_2 \longrightarrow CCl_3CHO + 3HCl$

$2CCl_3CHO + Ca(OH)_2 \longrightarrow 2CHCl_3 + (HCOO)_2Ca$

71. [A] i → f; ii → h; iii → a; iv → g; v → c; vi → b; vii → d; viii → e

[B] i → d; ii → a; iii → e; iv → c; v → b

72. (b)

73. (a)

74. (a)

75. (a, d)

76. (a, b, d)

77. Answer (d)

78. (d)

79. (a)

80. (a)

81. (a → r, t); (b → r, s); (c → p); (d → q)

82. (a → s, u); (b → p, w); (c → q, v); (d → r, x)

83. Explanation Based

84. Explanatory

85. Explanation Based

86. The acdidified solution of (x) gives a black sulphide with H_2S indicates that the chloride is of a basic cation of second group. The sulphide is insoluble in ammonium sulphide, hence, the cation belongs to IIA group of mixture analysis.

It gives white precipitate with $SnCl_2$ which turns to grey in excess of $SnCl_2$ suggests that (x) is $HgCl_2$. It is further confirmed by the reaction with KI.

87. (i) $2CuSO_4 + 2NaCl + SO_2 + 2H_2O \longrightarrow \underset{\text{Cuprous chloride}}{Cu_2Cl_2} + Na_2SO_4 + 2H_2SO_4$

(ii) $Ag + 2HNO_3\,(conc.) \longrightarrow AgNO_3 + NO_2 + H_2O$

(iii) $2AgNO_3 + Zn \longrightarrow 2Ag + Zn(NO_3)_2$

(iv) $HgCl_2 + SnCl_2 \longrightarrow Hg + SnCl_4$

(v) $ZnSO_4 + NaHCO_3 \longrightarrow ZnCO_3 + NaHSO_4$

$ZnCO_3 + 2HCl(dil.) \longrightarrow ZnCl_2 + H_2O + CO_2$

(vi) $2Fe + 3Cl_2 \longrightarrow 2FeCl_3$

(By passing chlorine over heated iron fillings)

(vii) $2FeSO_4 \xrightarrow[\text{strongly}]{\text{Heated}} Fe_2O_3 + SO_2 + SO_3$

$Fe_2O_3 + 6HCl \longrightarrow 2FeCl_3 + 3H_2O$

88. (i) On heating, it reacts with water of crystallisation and converted into Fe_2O_3 with evolution of HCl.

$2\left[FeCl_3 \cdot 6H_2O\right] \xrightarrow{\text{Heat}} Fe_2O_3 + 6HCl + 9H_2O$

(ii) ZnO is not blackened in atmosphere of hydrogen sulphide. It can be used both as oil and water paint. It is non-poisonous in nature.

(iii) Cu^{2+} ion has $3d^9$ configuration, i.e., one unpaired orbital while Zn^{2+} ion has $3d^{10}$ configuration, i.e., all orbitals doubly occupied.

(iv) On heating it reacts with water of crystallisation and forms zinc oxychloride.

$2ZnCl_2 \cdot 2H_2O \longrightarrow Zn_2OCl_2 + 2HCl + 3H_2O$

(v) Zinc coating protects the iron from corrosion as zinc is not affected by oxygen and water.

(vi) The oxide of copper which remains in the copper (blister) is reduced by introducing poles of green wood. The gas (hydrocarbons, CO, H_2, etc.) bubbles originating from the wood act as reducing agents. The mass is stirred with these green poles vigorously.

(vii) Hydrolysis of ferric chloride occurs giving finally brown precipitate.

$FeCl_3 + 3H_2O \rightleftharpoons Fe(OH)_3 + 3HCl$

$2Fe(OH)_3 \longrightarrow \underset{\text{Brown}}{Fe_2O_3} + 3H_2O$

(viii) Because of the availability of d-orbitals they can easily form intermediate products which are activated. The sizes of transition metal atoms and ions are also favourable for transitory complex formation with the reactants.

(ix) Copper in presence of moisture oxygen and carbon dioxide of atmosphere is converted into basic carbonate which is deposited as green layer on surface.

$2Cu + CO_2 + H_2O + O_2 \longrightarrow Cu(OH)_2CuCO_3$

89. (a) $K_2Cr_2O_7$ is not hygroscopic and can be used as primary standard.

(b) Mn^{2+}. It has maximum number of unpaired electrons.

(c) $SnCl_2$ is a strong reducing agent. It reduces $HgCl_2$ to Hg.

$2HgCl_2 + SnCl_2 \longrightarrow Hg_2Cl_2 + SnCl_4$

$Hg_2Cl_2 + SnCl_2 \longrightarrow 2Hg + SnCl_4$

(d) In the presence of organic matter and light, $AgNO_3$ decomposes to give a black stain of metallic silver.

$2AgNO_3 \longrightarrow 2Ag + 2NO_2 + O_2$

(e) In the solution, the following equilibria exists:

$Cr_2O_7^{2-} + H_2O \rightleftharpoons 2CrO_4^{2-} + 2H^+$

In acidic medium (pH < 7), it exists as $Cr_2O_7^{2-}$ ions and has orange colour while in basic medium (pH > 7), it exists as CrO_4^{2-} ions and has yellow colour.

90. Match the column

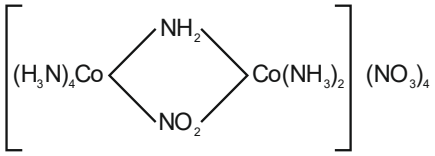
(A) (i → h); (ii → k); (iii → a); (iv → j); (v → l); (vi → i); (vii → b); (viii → d); (ix → c); (x → e); (xi → g); (xii → f)

(B) (i → e); (ii → l); (iii → g); (iv → f); (v → b); (vi → h); (vii → j); (viii → d); (ix → a); (x → c)

91. (b)

92. (a) $[Pt(NH_3)_2Cl(en)NO_2]Cl$

(b) $Ca_2[Fe(CN)_6]$

(c) 

(d) $[Co(CN)_2Cl_2]Cl_3$

(e) $Na[Al(OH)_4]$

(f) $K_2[Fe(EDTA)]$

(g) $[Cr(NH_3)_6][Co(CN_6)]$

93. (d)

94. (a)

95. (b)

96. (d)

97. (b)

98. (c)

99. (b)

100. Answer (a → p, s); (b → p, s); (c → q); (d → q, r)

101. Answer (a → p); (b → p); (c → q); (d → r); (e → s)

102. Answer (a → p, s); (b → q); (c → q, r); (d → p, s)

103. (i) Ammonium aqua pentafluorido ferrate (III)

(ii) Pentaammine aqua cobalt (III) chloride

(iii) Pentaammine aqua cobalt (III) tris oxalato cobaltate (III)

(iv) Pentaammine aqua cobalt (III) diammine tetrathiocyanato chromate (III)

(v) Pentaammine aqua cobalt (III) diammine tetranitrato cobaltate (III)

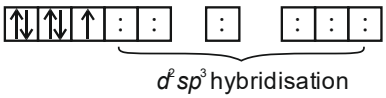
(vi) Potassium octacyanido molybdate (IV)

(vii) Pentaammine isothiocyanato chromium (III) tetrachloridozincate (II)

(viii) Bis thiourea bismuth (III) chloride

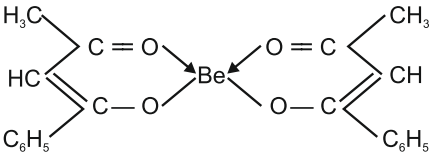
(ix) Tetramethyl ammonium tetra isothiocyanato cobaltate (II)

104. Octahedral complex has d^2sp^3 hybridisation



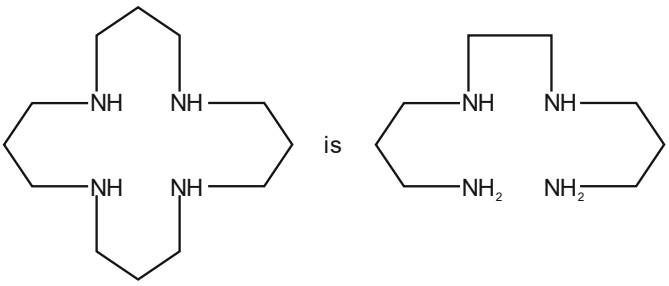
d^2sp^3 hybridisation

$[V(CO)_6] \Rightarrow$ Not show geometrical isomerism.

105.  and its mirror image

bis (Benzoyl acetonate) beryllium (II) complex

106. Thermodynamic stability of a complex formed with a cyclic polydentate ligand is increased when compared to the complex formed with a non-cyclic ligand.



107. $\Delta = h\nu = h.\frac{c}{\lambda}$ for one electron

or $\lambda = \frac{h \cdot c}{\Delta} = \frac{6.63 \times 10^{-34}\,Js \times 3 \times 10^{8}\,ms^{-1} \times 6.02 \times 10^{23}}{243 \times 10^{3}\,J}$

$= 492 \times 10^{-9}$ m = 492 nm, orange.

108. $[Cu(en)_2]^{2+}$ has two rings in the structure. On the other hand $[Fe(EDTA)]^-$ has five rings in the structure.

109. (d)

110. (c)

111. **Compound A :**

(i) Compound 'A' is ferrous sulphate which gives SO_2 and SO_3 on heating.

$$\underset{\text{Ferrous sulphate}}{2FeSO_4} \longrightarrow Fe_2O_3 + SO_2 \uparrow SO_3 \uparrow$$

Compound B :

(ii) and (iii) Compound 'B' is $BaCl_2$ which imparts green colour to flame and does not give any precipitate with H_2S

(iv) $BaCl_2$ forms chromyl chloride, when treated with $K_2Cr_2O_7$ and conc. H_2SO_4.

$$2BaCl_2 + K_2Cr_2O_7 + 3H_2SO_4 \longrightarrow K_2SO_4 + 2BaSO_4 + \underset{\substack{\text{Chromyl chloride}\\(\text{Red gas})}}{2CrO_2Cl_2} + 3H_2O$$

Chromyl chloride forms, yellow coloured Na_2CrO_4 with aqueous NaOH.

$$CrO_2Cl_2 + 4NaOH \longrightarrow \underset{\substack{\text{Yellow}\\(\text{Sodium chromate})}}{Na_2CrO_4} + 2NaCl + 2H_2O$$

112. Compound (X) should KI.

113. NaOH and $ZnSO_4$ are present in the solid.

114. $\underset{\substack{\text{Ferrous sulphide}\\\text{Black (A)}}}{FeS}$ $\underset{(B)}{H_2S}$ $\underset{(C)}{2HNO_3}$ $\underset{\substack{\text{Colloidal sulphur}\\(D)\,(\text{Turbid})}}{S}$ $\underset{(E)}{CuSO_4}$ $\underset{\substack{\text{Precipitate}\\(F)}}{CuS}$

115. (i) Gas given by heating mixture with NaOH gives brown ppt. with Nessler's reagent K_2HgI_4. It shows that the salt is a ammonium salt.

(ii) $BaCl_2$ solution gives white precipitate sparingly soluble in HCl. It shows the presence of SO_4^{2} ion in the mixture.

(iii) Blue colour with $K_3Fe(CN)_6$ indicates the presence of Fe^{2+} ion.

(iv) $\text{Mixture} + K_2Cr_2O_7 \xrightarrow[\Delta]{H_2SO_4} \underset{(A)}{\text{Red vapour}}\ (CrO_2Cl_2)$

It shows that the mixture contains Cl^- ion. Reactions involved may be given as

116. The compound (A) is a sulphate as its solution in dilute HCl on treatment with barium chloride solution gives a white precipitate of barium sulphate (F) which is insoluble in conc. HNO_3 and conc. HCl.

Solution of (A)

117. (A) Cr_2O_3; (B) Na_2CrO_4; (C) $Na_2Cr_2O_7$

118. (A) CaC_2O_4; (B) CaO; (C) CO; (D) CO_2

119. Reasoning Based

120. (b)

121. (b)

122. (b)

123. (a)

124. (b)

125. (a $\rightarrow$ p, q); (b $\rightarrow$ r, s); (c $\rightarrow$ p, s); (d $\rightarrow$ r, s)

126. (a $\rightarrow$ s); (b $\rightarrow$ p); (c $\rightarrow$ s); (d $\rightarrow$ q, r)

127. (SO_2)

128. (A $\rightarrow$ Pb_3O_4; B $\rightarrow$ PbO_2; C $\rightarrow$ $Pb(NO_3)_2$; D $\rightarrow$ PbS; E $\rightarrow$ $HMnO_4$)

129. (a $\rightarrow$ p, r); (b $\rightarrow$ q, s); (c $\rightarrow$ p, t); (d $\rightarrow$ p)



MATHEMATICS

ANSWERS

3. $-\frac{31}{10}$

4. $\sqrt{\frac{3}{2}}$

5. $\frac{13^{\frac{3}{16}}}{3^{\frac{3}{2}}}$

6. 384

7. $\frac{1}{10}$

8. $3 \cdot 5^{\frac{1}{5}}$

9. 0

10. $\frac{5-b}{2(a+ab+1-2b)}$

11. (i) $h(x) = 1, \quad x \geq 1$
$= x, \quad x < 1;$

(ii) $\phi(x) = x^2 + \frac{1}{x}, \quad x \geq 1$
$= x^2 + 1, \quad 0 \leq x < 1$
$= x + 1, \quad x < 0;$

(iii) $\psi(x) = x, \quad x \geq 1$
$= x^2, \quad 0 \leq x < 1$
$= x, \quad x < 0\}$

12. $g(x) = x + 4, \quad -4 \leq x < -3$
$= 0, \quad x = -3$
$= x + 2, \quad -3 < x < -1$
$= 0, \quad x = -1$
$= x, \quad -1 < x < 1$
$= 0, \quad x = 1$

13. 1

14. 16

15. $-\frac{1}{6}$

16. $\frac{34}{23}$

17. ∞

18. 0

19. 6

20. 0, $-\infty$

21. 0

22. 1

23. –1

24. $\frac{a+b+c}{3}$

25. $-|a|$

26. $\frac{a_1-b_1}{n}$

27. 0

28. $e^{-\frac{(n+1)(2n+1)}{12}}$

29. $\begin{Bmatrix} \frac{1}{2}, & a>0 \\ 0, & a=0 \\ -\frac{1}{2}, & a<0 \end{Bmatrix}$

30. $\begin{Bmatrix} 1, & a>1 \\ e, & a=1 \\ \infty, & a<1 \end{Bmatrix}$

31. $\frac{1}{2}$

32. $h(x) = -1, \quad x=(2n+1)\frac{\pi}{2}$

$= 0, \quad x \neq (2n+1)\frac{\pi}{2}, 0$

$= 1, \quad x = 0$

33. $a=\frac{1}{e}$

34. Discontinuous at $x = \pm 1$

37. 1

38. Continuous, not differentiable

39. $a = 1$

40. $\frac{1+x^4}{1+x^6}$

41. $\frac{x^2-1}{x^2-4}$

42. $\sqrt{\frac{a-x}{x-b}}$

43. $\frac{\sqrt{x^2+1}}{x}$

44. $\frac{1}{x^4+x^2+1}$

45. $8\left(\frac{4}{\pi^2+16}-\frac{1}{\ln 2}\right)$

49. $(-\infty, -3)$ decreasing, $\left(-3, \frac{1}{2}\right)$ increasing, $\left(\frac{1}{2}, 4\right)$ decreasing, $(4, \infty)$ increasing

50. $a \in \left(-\infty, -\sqrt{e^{11}+1}\right] \cup \left[\sqrt{e^{11}+1}, \infty\right)$

51. 120

52. (i) Concave upward in $(-\infty, -2)$; concave downward in $(-2, 2)$; concave upward in $(2, \infty)$.
    (ii) Concave upward in $(-\infty, \infty)$
    (iii) Concave downward in $(-\infty, 0)$; concave upward in $(0, \infty)$.
    (iv) Concave downward in $(-\infty, 0)$; concave upward in $(0, 1)$; concave upward in $(1, \infty)$.

53. $a = \frac{1}{6}$, $b = \frac{1}{2}$, $c = d = 1$

57. $n = \pm 1, \pm 3, \pm 5, \pm 15$

61. tan 20 – 20

64. Discontinuous and not differentiable at $x = \pm\sqrt{n}$, $n \in N$. Continuous and differentiable $\forall x$.

65. $a = \frac{I}{2} = 0, \pm\frac{1}{2}, \pm 1, \pm\frac{3}{2}, \ldots..$

66. 6

67. $g(x) = \left(\frac{x^{\frac{1}{3}}-b}{a}\right)^{\frac{1}{2}}$

69. One positive real root, one negative real root, two imaginary roots.

70. One root is 0, eight imaginary roots.

71. Six

72. $\left[-\frac{1}{2}\right] \cup \left[-\frac{2}{3}\right] \cup [3]$

73. $[1] \cup \left[-\frac{5}{2}\right]$

74. $[0] \cup [-1]$

75. [2]

76. $[-6] \cup [-5] \cup \left[-\frac{11}{2}\right]$

77. [0]

78. [81]

79. $[-1] \cup [1]$

80. 0

81. $[-2] \cup [2]$

82. $\left[-\frac{1}{4}\right]$

83. $\left[\sqrt{\frac{1+\sqrt{5}}{2}}\right]$

84. ϕ

85. $\left[\frac{1}{10}\right] \cup [2] \cup [1000]$

86. $\left(-\infty, -\sqrt{\log_2 \frac{3+\sqrt{5}}{2}}\right] \cup \left[\sqrt{\log_2 \frac{3+\sqrt{5}}{2}}, \infty\right)$

87. $[\log_{13} 5, 1]$

88. $[\log_5 7, 2]$

89. $\left(\frac{1+\sqrt{5}}{2}, 2\right)$

90. $a < -\frac{4}{5}$, $x \in \phi$

$a = 1$, $x \in \left[-\frac{7}{6}\right]$

$a = -\frac{4}{5}$, $x \in \left[-\frac{1}{3}\right]$

$a > -\frac{4}{5},\ a \neq 1, \quad x \in \left[\frac{-(2a+1)-\sqrt{5a+4}}{a-1}\right] \cup \left[\frac{-(2a+1)+\sqrt{5a+4}}{a-1}\right]$

91. $a = 0, \quad x \in \phi$

$a = 1, \quad x \in [-1]$

$a = -2, \quad x \in \left[\frac{1}{3}\right]$

$a \neq 0, 1, -2, \quad x \in [-1] \cup \left[\frac{a+1}{a-1}\right]$

93. (5, 3), (3, 5)

94. (8, 4), (4, 8)

95. (4, 2), $\left(\frac{26}{11}, \frac{52}{17}\right)$, (–52, 26), $\left(-\frac{26}{11}, \frac{52}{11}\right)$

96. $x \in R$, $y = 1$; (2, –3)

97. $\left(\frac{1}{3}, \frac{1}{4}\right)$

98. (1, 1), $\left(\frac{\sqrt{1+8n}-1}{2}, \frac{4n+1-\sqrt{1+8n}}{2}\right)$

99. (0, 0, 0), $\left(\frac{12}{5}, \frac{12}{7}, -12\right)$

100. (6, 5, 3), $\left(\frac{10}{3}, \frac{19}{3}, \frac{13}{3}\right)$

101. $\frac{7}{6}$

103. $x \in \left(0, \frac{1}{2}\right)$

104. 43

107. 0

108. $a \in \left[1-\sqrt{2}\right] \cup \left[5+\sqrt{10}\right]$

109. $a \in [2, \infty)$

110. 1539, 1541

767, 769, 771, 773

299, 301, ...10 numbers

207, 209, ...14 numbers

135, 137, ...20 numbers

119, 121, ...22 numbers

83, 85, ...28 numbers

27, 29, ...44 numbers

111. 25

112. 1540

114. $\frac{2\sqrt{3}}{2\sqrt{3}-\sqrt{2}+1}$

116. e^2

119. $\frac{n(n+1)(n+2)(3n+1)}{24}$

120. $\frac{n(n-1)(2n-1)}{3}$ yards

121. $\ln\frac{n+2}{2(n+1)}, \ -\ln 2$

125. $\frac{1}{(1+x)(1-x)^2}$

126. $f_n(x)=\frac{x-15}{3^n}+15$

127. 0

130. $\left(\frac{4-5t}{7}, \ \frac{13t-9}{7}, \ t\right)$

148. 1

149. $2x^2+2x\cos(A-B)-2$

154. $\left[\frac{n\pi}{2}+\frac{\pi}{4}\right]$

155. $\left[n\pi+\frac{\pi}{2}\right]\cup\left[k\pi+\tan^{-1}7\right]\cup\left[m\pi+\tan^{-1}3\right]$

156. ϕ

157. ϕ

158. $\left[n\pi+\frac{\pi}{2}\right]\cup\left[k\pi\pm\frac{1}{2}\cos^{-1}\left(-\frac{1}{4}\right)\right]$

159. $\left[n\pi+\frac{\pi}{4}\right]\cup\left[\frac{k\pi}{2}+(-1)^k\frac{1}{2}\sin^{-1}\left(\frac{\sqrt{5}-1}{2}\right)\right]$

160. $\left[\frac{n\pi}{4}\pm\frac{\pi}{12}\right]$

161. $\left[\frac{n\pi}{4}+\frac{\pi}{8}\right]$

162. $\left[\frac{n\pi}{4}+\frac{\pi}{8}\right]$

163. ϕ

164. $\left[n\pi-\tan^{-1}\frac{1}{6}\right]\cup\left[k\pi-\tan^{-1}\frac{1}{3}\right]$

165. $\left[2n\pi+\frac{\pi}{2}\right]$

166. $\left[n\pi\pm\frac{1}{2}\cos^{-1}\frac{1}{3}\right]\cup\left[k\pi\pm\frac{1}{2}\cos^{-1}\left(-\frac{1}{4}\right)\right]$

167. ϕ

168. $\left[n\pi+\frac{\pi}{2}\right]\cup\left[\frac{k\pi}{4}+\frac{\pi}{8}\right]$

169. $[4n\pi + \pi]$

170. $[24n\pi + 2\pi]$

171. $\left[n\pm\frac{1}{6}\right]$

172. $\left[n\pi+\frac{\pi}{2}\right]\cup\left[k\pi\pm\tan^{-1}\frac{1}{2}\right]$

175. $[0]\cup\left[\frac{1}{2}\right]$

176. 0

177. $\left[2n\pi-\frac{\pi}{3},\ 2n\pi\right)\cup\left[2k\pi+\frac{\pi}{3},\ 2k\pi+\pi\right)$

178. $\left(2n\pi+2\cot^{-1}2,\ 2n\pi+2\cot^{-1}\left(-\frac{1}{2}\right)\right)$

179. $\left(n\pi-\frac{1}{2}\cos^{-1}\frac{1}{3},\ n\pi-\frac{\pi}{6}\right)\cup\left(k\pi,\ k\pi+\frac{\pi}{6}\right)\cup\left(m\pi+\frac{1}{2}\cos^{-1}\frac{1}{3},\ m\pi+\frac{\pi}{4}\right)\cup\left(p\pi+\frac{2\pi}{3},\ p\pi+\frac{3\pi}{4}\right)\cup\left(\ell\pi+\frac{\pi}{3},\ \ell\pi+\frac{\pi}{2}\right)$

180. $\left(\frac{2n\pi}{5}-\frac{\pi}{10},\ \frac{2n\pi}{5}-\frac{\pi}{30}\right)\cup\left(\frac{2k\pi}{5}+\frac{\pi}{10},\ \frac{2k\pi}{5}+\frac{7\pi}{30}\right)$

181. $\left(n\pi-\frac{\pi}{3},\ n\pi-\frac{\pi}{9}\right)\cup\left(k\pi,\ k\pi+\frac{2\pi}{9}\right)\cup\left(m\pi+\frac{\pi}{2},\ m\pi+\frac{5\pi}{9}\right)$

182. $\left[n\pi-\frac{7\pi}{12},\ n\pi-\frac{\pi}{2}\right)\cup\left(k\pi-\frac{\pi}{2},\ k\pi+\frac{\pi}{12}\right]$

183. $\left(\frac{1}{\sqrt{2}},\ 1\right)\cup\left(-1,\ -\frac{1}{\sqrt{2}}\right)$

184. $\frac{1}{2}(\tan 3^n\theta - \tan\theta)$

186. $\left(0, \frac{\pi}{2}\right)$ decreasing, $\left(\frac{\pi}{2}, \frac{2\pi}{3}\right)$ increasing, $\left(\frac{2\pi}{3}, \pi\right)$ decreasing.

193. $\frac{2}{5}$

199. $\frac{313}{338}$

223. $\left(\frac{2}{\sqrt{3}} \pm 1\right)$

226. $\frac{2}{\sqrt{3}}, \frac{1}{\sqrt{6}}$

❑ ❑ ❑